INCH-POUND
A-A-59193
20 May 1998
SUPERSEDING
MIL-B-2678C
4 August 1989

COMMERCIAL ITEM DESCRIPTION

BRUSHES, WIRE, BOILER TUBE, EXPANDING TYPE FOR USE WITH ELECTRIC AND AIR MOTOR DRIVEN BOILER TUBE CLEANING OUTFITS

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1.1 This CID covers four types of expanding wire brushes and brush refills for use with air or electric motor driven boiler tube cleaning outfits.

2. CLASSIFICATION

2.1 The brushes shall be of the following classes:

Class A - For use with electric motor driven tube cleaning outfits (2 inches or smaller) (Army and Air Force use only).
Class B - For use with air motor driven boiler tube cleaning outfits (2 inches or smaller).
Class C - Integral Universal Joint type. For use with either electric or air motor driven boiler tube cleaning outfits.
Class D - Large expanding brushes (Greater than 2 inches). For use with either electric or air motor driven boiler tube cleaning outfits.

Beneficial comments recommendations, additions, deletions, clarifications etc. and any other data which may improve this document should be sent to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160.

AMSC N/A FSC 5120

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3. SALIENT CHARACTERISTICS

3.1 Brushes shall expand and contract to provide automatic adjustment of the outside diameter of brushes to accommodate variations in the inside diameter of straight and curved sections of boiler tubes.

3.2 Tube dimensions. Size of brushes and the degree of expansion and contraction provided shall be such that, when operated with the corresponding type and size of boiler-tube cleaning outfit per A-A-59192, the brushes shall pass through and effectively operate in straight and curved sections of sizes of boiler tubes with dimensions shown in table I.

3.3 Bodies of brushes. Brushes shall be constructed so that the bristle sections can be renewed without renewing the brush body. Bodies of brushes shall conform to the dimensions shown on figure 1 so that refills (replacement bristle sections) can be interchanged regardless of their manufacturer. All parts of brushes shall be made of steel. Working parts exposed to wear shall not exceed a maximum hardness of Rockwell B-68 or a minimum of Rockwell B-60 when tested in accordance with ASTM E 18.

3.4 Bristles. Only high tempered carbon steel wire bristles shall be used in brushes intended for removing deposits on the interior of boiler tubes. The chemical composition, tensile strength, corrosion resistant coating, and wire diameter shall be in accordance with the best commercial practice to provide both effective cleaning and long life (5 hours cleaning dirty boiler tubes). Bristle ends shall not be trimmed by grinding which will soften the wire and prevent brushes from meeting the 5 hour minimum life expectancy.

3.5 Threaded connections. Bodies of brushes shall be threaded for connection with flexible brush holders that conform to the requirements shown in table II. Thread dimensions, tolerances, and gauges for determination of interchangeability, shall conform to FED-STD-H28. Unified coarse thread (UNC) series 2A-2B shall be the standard for screw threading; UNC series 3A-3B standards are also acceptable.

3.6 Required brush dimensions and features.

3.6.1 Class A. Brushes of size 2 inches or smaller for use with electric motor driven tube cleaning outfits conforming to A-A-59192. Brushes shall conform to the dimensions shown in table III. Each brush shall be provided with mechanical means, independent of and in addition to the centrifugal force caused by rotation of the brush, for automatic adjustment of the outside diameter of the brush variations in the inside diameter of the tubes. Not for shipboard use.

3.6.2 Class B. Brushes of size 2 inches or smaller for use with air motor driven tube cleaning outfits conforming to A-A-59192. Brushes shall conform to the dimensions shown in table IV.

3.6.3 Class C. Brushes of sizes 2 inches or smaller with integral universal joint for use with either electric or air motor driven tube cleaning outfits conforming to A-A-59192. The brushes shall conform to the dimensions shown in table V. The universal joint shall be attached to the end of the brush that joins the brush holder. The joint design must permit operation in tubes bent to a radius of 2-3/4 inches through 180 degrees where the crush of tube does not exceed 1.5 times that shown in table I.

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3.6.4 Class D. Brushes of sizes larger than 2 inches, up to and including 5 inches, for use with electric or air motor driven tube cleaning outfits. The brushes shall conform to the dimensions shown in table VI.

4. REGULATORY REQUIREMENTS

4.1 The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability.

5.2.1 A market tested item is required to assure reliability and cleaning effectiveness without costly performance testing. The company producing the item must have at least 5 years experience designing and producing pneumatic and electric boiler tube cleaning equipment. Service proven designs and established quality assurance programs are essential to minimize brushes lodging in boiler tubes or otherwise damaging tubes so that costly boiler repairs will be necessary.

5.2.2 Test and inspection. Bid samples may be required when necessary to assure brush and brush refill set quality and interchangeability; as well as, brush compatibility with A-A-59192 tube cleaning outfits.

6. PACKAGING

6.1 Packaging shall meet the standard practice requirements of ASTM D 3951 unless special preservation, packaging, packing, labeling, and marking are specified in the contract or order.

7. NOTES

7.1 CID-based parts identification number. The following part identification number procedure is for government purposes and does not constitute a requirement for the contractor. This example describes a part numbering system for A-A-59193.

Example: AAXXXXB1608B262

Class B brush for 2" OD tube
with 08 gage (BWG)(0.165 " wall);
7/16-18 UNC male thd.
0.262 min free bristle length

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Class designation	=	A,B,C,or D
Tube OD in eighths of an inch	=	8 to 40
Tube wall thickness in BWG	=	00 to 20
Body threads :		
5/16-18M	=	A
7/16-14M	=	B
9/16-12F	=	C
5/8-11M	=	D
5/8-11F	=	E
3/4-10F	=	F
7/8-09F	=	G
Free bristle length in thousandths of an inch (minimum allowed)	=	076 to 662

7.2 Source documents.

7.2.1 ASTM standards are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2859.

7.2.2 Federal Standard FED-STD-H28 may be obtained by sending a request for copies to the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.

7.3 National Stock Numbers (NSNs). The following is a list of NSNs associated with electric and pneumatic motor driven brush assemblies and brush refill sets (replacement bristle sets). The list may not be indicative of all possible NSNs that could be associated with the CID.

Class A

Size	Brush NSN	Refill NSN	Size	Brush NSN	Refill NSN
1 x 0.085/ .095	5130-00-278-1284	-	1-1/2 x 0.120	5130-00-278-1292	-
1 x 0.109	5130-00-278-1285	-	1-3/4 x 0.109	5977-01-134-2469	5130-01-132-9527
1-1/8 x 0.120	5130-00-278-1287	-	1-3/4 x 0.134	5130-00-278-1295	5130-00-260-5994
1-1/4 x 0.109	5130-00-278-1288	-	2 x 0.134	5130-00-260-8237	5340-01-132-9527
1-1/4 x 0.134	5130-00-278-1289	-	2 x 0.165	5130-00-260-8238	-

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Class B

Size	Brush NSN	Refill NSN	Size	Brush NSN	Refill NSN
1 x 0.085	5130-00-277-3075	5130-00-779-5035	1-3/8 x 0.134	5130-00-277-3082	-
1 x 0.095	-	5130-00-260-6025	1-3/8 x 0.148	5130-00-165-3981	5130-00-165-3982
1 x 0.109	5130-00-277-3076	5130-00-260-6026	1-1/2 x 0.109	5130-00-177-7070	5130-00-177-7072
1 x 0.134	5130-00-062-1182	5130-00-062-6650	1-1/2 x 0.120	5130-00-277-3083	5130-00-260-6033
1-1/8 x 0.109	5130-00-277-3077	-	1-1/2 x 0.134	5130-00-277-3084	5130-00-260-6034
1-1/8 x 0.120	5130-00-277-3078	5130-00-260-6028	1-3/4 x 0.134	5130-00-277-3086	-
1-1/4 x 0.095	5130-00-177-7069	5130-00-177-7071	2 x 0.134	5130-00-277-3088	5130-00-260-6038
1-1/4 x 0.109	5130-00-277-3079	5130-00-260-6029	2 x 0.148	5130-00-062-1184	5130-00-062-1194
1-1/4 x 0.120	5130-00-062-1183	5130-00-062-1193	2 x 0.165	5130-00-174-3224	5130-00-260-6039
1-1/4 x 0.134	5130-00-277-3080	-	2 x 0.180	5130-00-062-1185	5130-00-062-1195
1-3/8 x 0.120	5130-00-277-3081	-	2-1/4 x 0.120	5130-01-059-4120	5130-01-059-4100

Class C - Universal joint

Size	Brush NSN	Refill NSN	Size	Brush NSN	Refill NSN
1 x 0.095	5130-00-277-3089	5130-00-260-6040	1-3/8 x 0.109	-	5130-00-900-3609
1 x 0.109	5130-00-277-3090	5130-00-260-6041	1-3/8 x 0.134	5130-00-277-3094	-
1 x 0.134	5130-00-062-1186	5130-00-062-1196	1-1/2 x 0.120	5130-00-174-3225	5130-00-260-6048
1-1/8 x 0.109	5130-00-277-3091	5130-00-260-6042	1-1/2 x 0.134	5130-00-174-3226	5130-00-260-6049
1-1/8 x 0.120	5130-00-062-1187	5130-00-260-6043	1-1/2 x 0.148	5130-00-177-7275	5130-00-177-7276
1-1/4 x 0.095	5130-00-775-0336	5130-00-779-5036	2 x 0.134	5130-00-277-3095	5130-00-260-6053
1-1/4 x 0.109	5130-00-277-3092	5130-00-260-6044	2 x 0.148	5130-00-062-1188	5130-00-062-1198
1-1/4 x 0.134	5130-00-277-3093	5130-00-260-6045	2 x 0.165	5130-00-277-3096	5130-00-260-6054
			2 x 0.180	-	5130-00-006-3413

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Class D - Sizes Larger than 2"

Size	Brush NSN	Refill NSN	Size	Brush NSN	Refill NSN
2-1/2 x 0.120	5130-00-260-8246	-	3-1/2 x 0.134	5130-00-260-8260	-
2-1/2 x 0.134	5130-00-260-8247	5130-00-260-5999	3-1/2 x 0.148	5130-01-059-4123	5130-01-059-4104
2-1/2 x 0.148	5130-00-260-8248	5130-00-260-6004	3-1/2 x 0.165	5130-00-260-8258	5130-00-260-6013
2-3/4 x 0.120	5130-00-260-8250	5130-01-059-4103	4 x 0.134	5130-00-260-8262	-
2-3/4 x 0.134	5130-01-059-4121	5130-01-059-4102	4 x 0.165	5130-00-260-8263	5130-00-260-6015
2-3/4 x 0.180	5130-01-059-4122	5130-01-059-4101	4 x 0.203	5130-00-260-8264	5130-00-260-6016
3 x 0.134	5130-00-260-8254	5130-00-260-6006	4 x 0.250	5130-00-260-8265	5130-00-260-6017
3 x 0.165	5130-00-260-8255	5130-00-260-6007	4-1/2 x 0.220	5130-00-260-8241	-
3 x 0.250	5130-00-260-8256	-	4-1/2 x 0.250	5130-00-260-8242	5130-00-260-6021
3-1/4 x 0.134	5130-00-260-8257	5130-00-260-6012	4-1/2 x 0.280	5130-00-260-8243	5130-00-260-6022
3-1/4 x 0.165	5130-00-260-8258	5130-00-260-6010	4-1/2 x 0.300	5130-00-837-2171	-
3-1/4 x 0.250	5130-00-260-8259	5130-00-260-6011	4-1/2 x 0.343	5130-00-260-8245	5130-00-260-6024
			5 x 0.300 pipe	5130-00-062-1191	5130-00-062-1201

7.4 Source of supply. These products are currently available commercially as shown below:

Vendors

Elliott Co.
Industrial Tool Division
1760 Tuttle Ave.
Dayton, OH 45401

Thomas C. Wilson Co.
21-11 44th Ave.
Long Island City, NY 11101-5088

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TABLE I. Dimensions of curved boiler tube sections.

Size of brush and outside diameter of boiler tube	Arc radius	Arc length	Maximum crush of tube
(inches)	(inches)	(degrees)	(inch)
1	5	75	0.050
1-1/8	5	75	0.050
1-1/4	5	75	0.062
1-3/8	6	75	0.062
1-1/2	6	75	0.075
1-5/8	6	75	0.075
1-3/4	6	75	0.100
2	6	75	0.120
2-1/2	12	75	0.140
2-3/4	12	75	0.165
3 to 4	12	75	0.190
4-1/2	14	75	0.215
5	16	75	0.240

TABLE II. Threading of bodies of brushes for connection with flexible brush holders.

Class of brush	Size of brush (inches)	Size of thread (inch)	Threads per inch (number)	Length of threaded section (inch)	Male or female
A, B, and C	1 to 1-1/2	5/16	18	5/16	Male
A, B, and C	1-5/8 to 1-3/4	7/16	14	3/8	Male
A	2	9/16	12	1/2	Female
B and C	2	7/16	14	3/8	Male
D	2-1/2 to 4	9/16	12	1/2	Female
D	4-1/2 to 5	7/8	9	9/16	Female

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Brush size (inches)	A (inch)	B (inches)	C (inches) (+/- 0.005)	D (inch) (+0.003/ -.000)	Number of slots
Class A - Sizes up to and incl. 2 inches					
1 and 1-1/8	3/32	9/16	1-15/16	0.159	6
1-1/4 and 1-3/8	3/32	11/16	1-11/16	0.222	6
1-1/2	3/32	13/16	1-13/32	0.222	6
1-5/8 and 1-3/4	1/8	15/16	1-27/64	0.224	6
2	3/16	1-3/16	1-5/8	0.224	8
Class B - Sizes up to and incl. 2 inches					
1 and 1/8	1/16	9/16	63/64	0.222	4
1-1/4 and 1-3/8	5/64	11/16	63/64	0.222	4
1-1/2	11/64	13/16	1-7/32	0.222	6
1-5/8 and 1-3/4	3/16	15/16	1-27/64	0.224	6
2	3/8	1-3/16	1-5/8	0.224	8

FIGURE 1. Boiler tube brush, expanding type body.

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Brush size (inches)	A (inch)	B (inches)	C (inches) (+/- 0.005)	D (inch) (+0.003/-.000)	Number of slots
Class C - Sizes up to and incl. 2 inches					
1 and 1-1/8	1/16	9/16	51/64	0.222	4
1-1/4 and 1-3/8	9/64	3/4	51/64	0.222	4
1-1/2	3/16	7/8	63/64	0.222	6
1-5/8 and 1-3/4	1/4	1	63/64	0.222	6
2	9/32	1-1/8	1-11/64	0.222	8
Brush size (inches)	A (inch)	B (inches)	C (inches) (+/- 0.005)	D (inch) (+0.005/-.000)	slots
Class D - Sizes larger than 2 inches up to and incl. 5 inches					
2-1/2	1/2	1-9/16	1-51/64	0.442	6
3 and 3-1/4	5/8	2	2-3/32	0.570	6
3-1/2 and 4	3/4	2-3/8	2-3/32	0.570	6
4 and 4-1/2	1-9/16	3	2-1/8	0.570	6
5	1-9/32	3-3/8	2-1/8	0.570	6

FIGURE 1. Boiler tube brush, expanding type body - Continued.

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TABLE III. Dimensions of class A brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.820	0.733	0.120	3360
1	0.109	0.782	0.792	0.705	0.106	3360
1-1/8	0.109	0.907	0.917	0.830	0.169	3360
1-1/8	0.120	0.885	0.895	0.808	0.158	3360
1-1/4	0.109	1.032	1.042	0.920	0.178	4800
1-1/4	0.134	0.982	0.992	0.870	0.153	4800
1-3/8	0.120	1.135	1.145	1.022	0.229	4800
1-3/8	0.134	1.107	1.117	0.994	0.215	4800
1-1/2	0.120	1.260	1.270	1.110	0.243	5760
1-1/2	0.134	1.232	1.242	1.082	0.229	5760
1-5/8	0.134	1.357	1.367	1.172	0.227	5760
1-3/4	0.134	1.482	1.492	1.298	0.290	5760
1-3/4	0.156	1.438	1.448	1.254	0.268	5760
2	0.134	1.732	1.742	1.542	0.287	8960
2	0.165	1.670	1.680	1.480	0.257	8960

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TABLE IV. Dimensions of class B brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.830	0.689	0.118	2560
1	0.109	0.782	0.802	0.660	0.104	2560
1	0.134	0.732	0.751	0.610	0.076	2560
1-1/8	0.109	0.907	0.927	0.786	0.167	2560
1-1/8	0.120	0.885	0.905	0.764	0.156	2560
1-1/4	0.109	1.032	1.052	0.894	0.198	2560
1-1/4	0.120	0.010	1.030	0.872	0.186	2560
1-1/4	0.134	0.982	1.002	0.844	0.173	2560
1-3/8	0.120	1.135	1.155	0.996	0.249	2560
1-3/8	0.134	1.107	1.127	0.968	0.235	2560
1-1/2	0.120	1.260	1.280	1.120	0.248	4800
1-1/2	0.134	1.232	1.252	1.092	0.234	4800
1-5/8	0.134	1.357	1.377	1.176	0.232	5760
1-3/4	0.134	1.482	1.502	1.302	0.295	5760
1-3/4	0.156	1.438	1.458	1.258	0.273	5760
2	0.134	1.732	1.752	1.522	0.292	8960
2	0.148	1.704	1.724	1.494	0.277	8960
2	0.165	1.670	1.690	1.460	0.262	8960
2	0.180	1.640	1.660	1.430	0.247	8960

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TABLE V. Dimensions of class C brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.810	0.705	0.125	1920
1	0.109	0.782	0.782	0.677	0.110	1920
1	0.134	0.732	0.732	0.627	0.080	1920
1-1/8	0.109	0.907	0.907	0.802	0.173	1920
1-1/8	0.120	0.885	0.885	0.780	0.162	1920
1-1/4	0.109	1.032	1.042	0.956	0.163	1920
1-1/4	0.134	0.982	0.992	0.906	0.138	1920
1-3/8	0.120	1.135	1.145	1.059	0.214	1920
1-3/8	0.134	1.107	1.117	0.031	0.200	1920
1-1/2	0.120	1.260	1.270	1.070	0.220	3840
1-1/2	0.134	1.232	1.242	1.042	0.206	3840
1-5/8	0.134	1.357	1.367	1.125	0.205	3840
1-3/4	0.134	1.482	1.492	1.246	0.240	3840
1-3/4	0.156	1.438	1.448	1.200	0.220	3840
2	0.134	1.732	1.742	1.422	0.321	6400
2	0.148	1.704	1.714	1.394	0.305	6400
2	0.165	1.670	1.680	1.360	0.290	6400

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TABLE VI. Dimensions of class D brushes (sizes larger than 2 inches, up to and including 5 inches).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) number
			Expanded (min) (inches)	Contracted length of bristle (min) (inch)		
2-1/2	0.120	2.260	2.369	1.924	0.307	1800
2-1/2	0.134	2.232	2.341	1.896	0.293	1800
2-1/2	0.148	2.204	2.313	1.868	0.279	1800
2-1/2	0.165	2.170	2.279	1.834	0.262	1800
2-3/4	0.120	2.510	2.519	2.174	0.432	1800
2-3/4	0.134	2.482	2.591	2.146	0.418	1800
2-3/4	0.148	2.454	2.563	2.118	0.404	1800
2-3/4	0.165	2.420	2.529	2.084	0.387	1800
3	0.134	2.732	2.825	2.417	0.391	2520
3	0.165	2.670	2.763	2.355	0.360	2520
3	0.250	2.500	2.667	2.259	0.312	2520
3-1/4	0.134	2.982	3.075	2.667	0.516	2520
3-1/4	0.165	2.920	3.013	2.605	0.485	2520
3-1/4	0.250	2.750	2.843	2.435	0.400	2520
3-1/2	0.120	3.260	3.353	2.902	0.428	2520
3-1/2	0.134	3.232	3.325	2.874	0.412	2520
3-1/2	0.165	3.170	3.263	2.812	0.381	2520
3-1/2	0.180	3.140	3.233	2.782	0.365	2520
4	0.134	3.732	3.825	3.374	0.662	2520
4	0.165	3.670	3.763	3.312	0.631	2520
4	0.203	3.594	3.687	3.236	0.593	2520
4	0.250	3.500	3.593	3.142	0.546	2520
4-1/2	0.180	4.140	4.233	3.729	0.578	2520
4-1/2	0.203	4.094	4.187	3.683	0.555	2520
4-1/2	0.220	4.060	4.153	3.649	0.538	2520
4-1/2	0.250	4.000	4.093	3.589	0.508	2520
4-1/2	0.280	3.940	4.033	3.529	0.478	2520
4-1/2	0.293	3.914	4.007	3.503	0.465	2520
4-1/2	0.343	3.814	3.907	3.403	0.415	2520
5	0.300	4.400	4.493	3.989	0.415	2520

MILITARY INTERESTS:

Custodians:
Army - G1
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 5120-0083)

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